

ANSHUL RAMAN

📍 Cincinnati, OH • 📞 (513) 383-2008 • ✉️ ramanal@mail.uc.edu

🌐 www.linkedin.com/in/anshulraman06/ | 🐙 www.github.com/itsramananshul | 🌐 anshulraman.com/



Education

University of Cincinnati — College of Engineering and Applied Science

B.S. Computer Science | GPA: 3.68 | Dean's List, CEAS

Cincinnati, OH

Expected May 2030

- **Relevant Coursework:** Computer Science I, Calculus I-II, Engineering Applications I-II
- **Activities:** IEEE, UC Robotics, AI Society, Bearcat Tech Innovators, Tennis Club

Experience

Founding Engineer, OpenPrem Cincinnati, OH

April 2026 – Present

- Architected and shipped **Relix**, a 496k-LOC, 14-crate Rust agentic control plane with a local-first single-binary deployment, ~90-route Axum API, SQLite state layer, React dashboard, and 46 versioned Windows releases.
- Built **Prime**, a governed AI-agent runtime with bounded think-tool-observe loops, strict JSON decision envelopes, fail-closed tool validation, approval gates, and self-correction for safe tool-using automation.
- Built **SolFlow**, a TypeScript visual workflow IDE backed by a Rust SOL compiler and bytecode VM compiled to WebAssembly, enabling graph-based editing, real diagnostics, source tracing, in-browser execution, & 407 automated tests.
- Shipped OpenPrem's enterprise demo layer, including OIDC/SAML SSO, TOTP 2FA, RBAC, SSE monitoring, an LLM assistant, a 15-deployment SaaS demo fleet, and a real-time mesh operations console.

Projects

ARIA — Ambient Reasoning & Intelligence Architecture

March 2026 – Present

Always-on personal AI system | Node.js, Next.js, FastAPI, Supabase, Qdrant, Gemini, Playwright, ESP32

- Built **ARIA**, a solo personal-AI system that unifies chat, Telegram, Gmail, Calendar, browser history, captured AI conversations, and a custom ESP32 voice device through one shared memory and OIDC login.
- Engineered a hybrid RAG memory layer over a 1,000+ note GitHub vault using Qdrant, Voyage embeddings, semantic + keyword retrieval, reranking, and SHA-diff indexing to recall context without reprocessing unchanged files.
- Developed an agentic execution layer using Gemini function-calling, bounded ReAct loops, Playwright browser automation, background jobs, and authenticated delegation to a Hermes code-execution agent.
- Converted a SenseCAP ESP32-S3 wearable into a low-latency voice and vision assistant with wake word, camera capture, Gemini STT, Gemini Live API TTS, Supabase device sync, and ~1–2s first-audio latency.

VoxVitals — AI Clinical Trial Platform | Next.js, Supabase, Gemini, ElevenLabs, Presage SDK

March 2026

- Led product architecture and full-stack development of a clinical trial monitoring system in 24 hrs at **Revolution UC 2026**, winning **3rd Place** (Medpace Clinical Trial Challenge) & Best Use of Presage API (MLH).
- Built AI-driven risk stratification, real-time patient timelines, dropout prediction, and doctor performance analytics; integrated Presage SDK for passive physiological sensing, ElevenLabs for voice check-ins & biometric phrase login, and Gemini for intelligent alert detection across a role-based multi-user system.

Gradintel | Python, Next.js, Supabase, Gemini, Grok, Claude, OpenAI

Feb 2026 – Mar 2026

- Shipped a full-stack GPA tracking platform to live users — a Python core for GPA calculation, grade simulation, weighted scoring, and at-risk detection; 4 LLM integrations (Gemini, Grok, Claude, OpenAI); Supabase cloud sync, gamification (XP/badges/streaks), data export, and GPA card sharing.
- Architected the platform end-to-end from data models and backend logic to UX flows, enabling students to simulate grade outcomes, pinpoint weak courses, sync progress across devices, and share mobile-responsive GPA reports.

Technical Skills

Languages: Python, TypeScript, JavaScript, Rust, C++, SQL, HTML, CSS

Frameworks & Tools: React, Next.js, Vue 3, Node.js, Express, FastAPI, Axum (Tokio), Supabase, Docker, Vercel, Git, WebAssembly

AI / ML: LLM orchestration (Gemini, OpenAI, Anthropic), RAG pipelines, Qdrant, MCP, agentic systems, wasm-bindgen

Awards & Activities

Gold Medal — National-Level Science Olympiad (2023)

Gold Medal — National Abacus Competition (2023)

Top 150 National — All India Tennis Association (AITA, 2022)

Volunteer: Mary Magdalen House (clothing intake); Bearcat Recycling (event waste sorting); Epic House (art for elderly residents)

Available for Co-op: Fall 2026